

VistaCap

**Visual
Intelligence
Smart Cap**



VISTACAP

Project by :
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**Project
Synopsis**



Who It's For



- People walking through busy streets
- Travelers reading signs in new languages
- Anyone who wants smart, hands-free assistance
- Folks communicating with sign language users
- Students, cyclists, explorers—you name it!



What It Can Do ?

- 👁 Detect objects, terrain, and gestures in real time
- 🗣 Recognize ISL/ASL sign language and convert it to speech or text
- 🌐 Translate printed text and spoken words instantly
- 🔊 Give audio alerts for obstacles and directions
- 🏠 Understand scenes—like emotions, crowds, or movement
- 🚗 Even support self-driving features when mounted on a vehicle





What's Inside the Cap



- 🧠 Raspberry Pi Zero 2 W (the brain)
- 📷 Camera module (your digital eyes)
- 🔊 Normal speaker (for voice and alerts)
- 💻 Display (shows text and translations)
- 🔋 Battery pack (portable power)



How It Works



- Uses AI models like YOLOv8n and MediaPipe
- Reads signs and menus with OCR
- Converts speech to text using Whisper or similar
- Translates languages using online APIs
- Gives feedback through sound, display, or even vibration



Where You Can Use It

-   Walking through crowded areas
-  Traveling in new cities
-  Learning or teaching in classrooms
-   Cycling with awareness
-  Supporting focus and sensory clarity



Thank You



VISTA CAP



PROJECT WEBSITE :

<https://more.criusbehera.com/vista-cap/>



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